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(19) **United States**(12) **Patent Application Publication**
BAEK et al.(10) **Pub. No.: US 2025/0277188 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **NUCLEATED CELL DEATH INDUCTION
METHOD USING LOW TEMPERATURE
TREATMENT****Publication Classification**(51) **Int. Cl.***C12N 5/078* (2010.01)*C12N 5/00* (2006.01)(52) **U.S. Cl.**CPC *C12N 5/0641* (2013.01); *C12N 5/0087*
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2523/00 (2013.01); *C12N 2529/10* (2013.01)(71) Applicant: **ARTBLOOD INC.**, Seoul (KR)(72) Inventors: **Eun Jung BAEK**, Seoul (KR); **Suyeon KIM**, Seoul (KR)(21) Appl. No.: **18/850,298**(22) PCT Filed: **Apr. 12, 2023**(86) PCT No.: **PCT/KR2023/004918**

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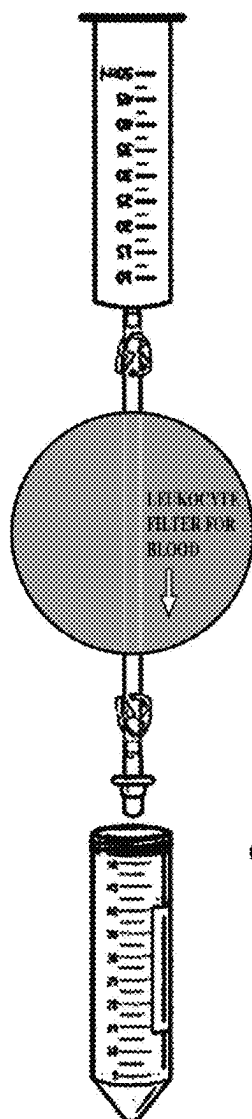
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(57)

ABSTRACT

The present invention relates to a nucleated cell death induction method in which a low temperature treatment is performed on incubating blood cells such that only nucleated cells are damaged and enucleated blood cells are not damaged. According to the present invention, nucleated cell death can be induced by means of a simple, cost-effective and safe method.

Specification includes a Sequence Listing.

Wash cells with 1x PBS, and then suspend cells in 30 mL 1x PBS



After descending by gravity, wash once more with 30 mL 1x PBS

Collect cells



Centrifuge at 4,000 xg for 30 minutes at 4°C



Carefully remove supernatant while leaving about 1 mL, and then transfer to 1.5 mL tube



Centrifuge at 12,000 xg for 1 minute



Carefully remove supernatant while leaving 50 to 100 µL



Genomic DNA extraction or Wright-Giemsa staining

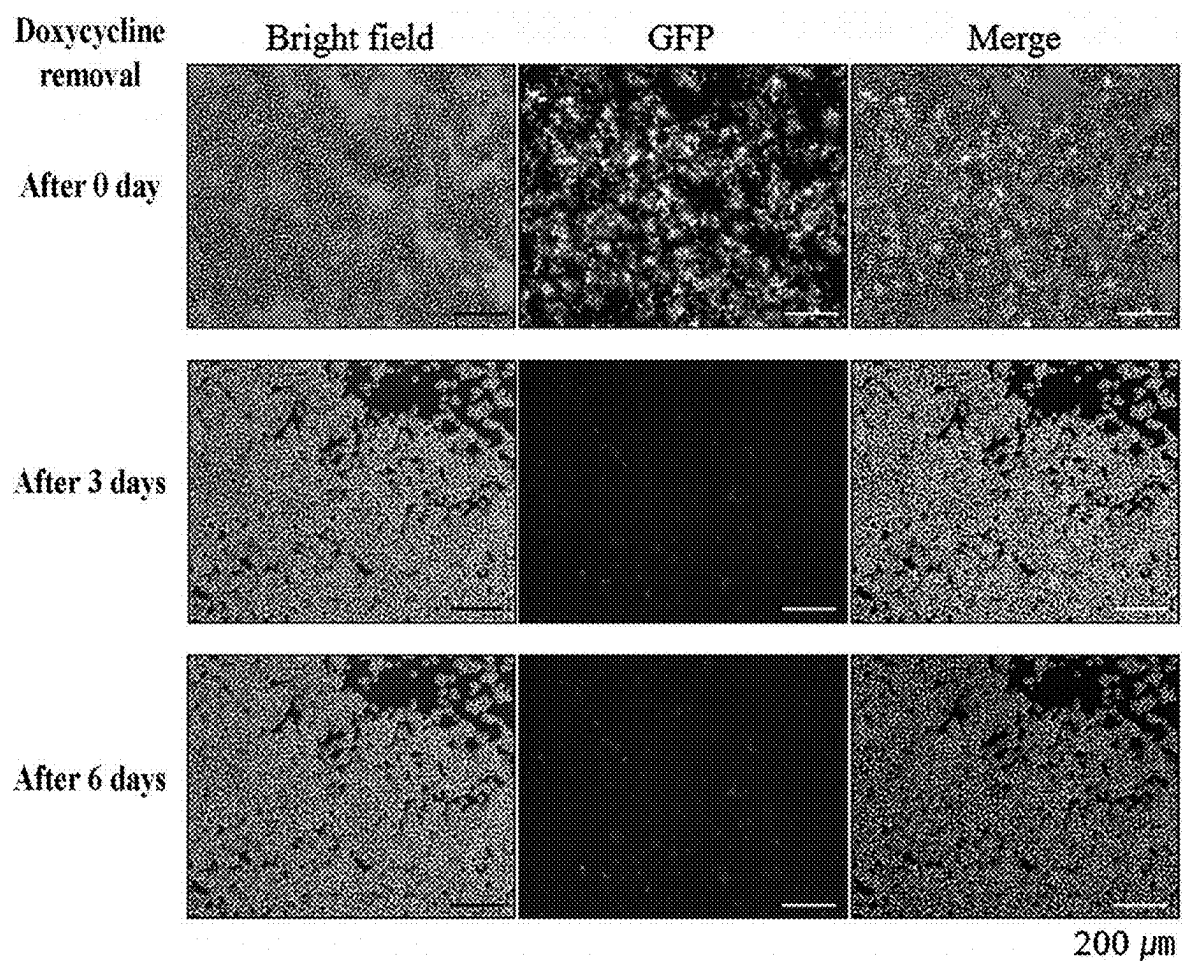


FIG. 1A

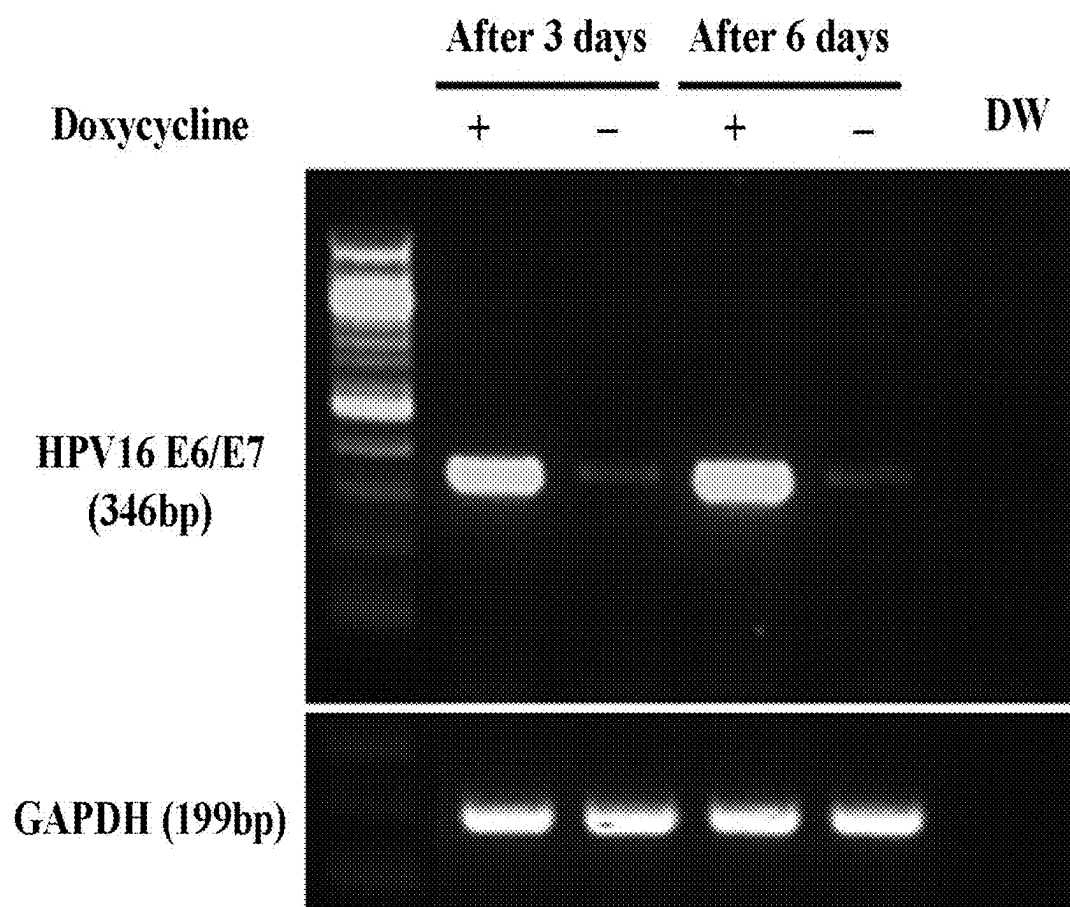


FIG. 1B

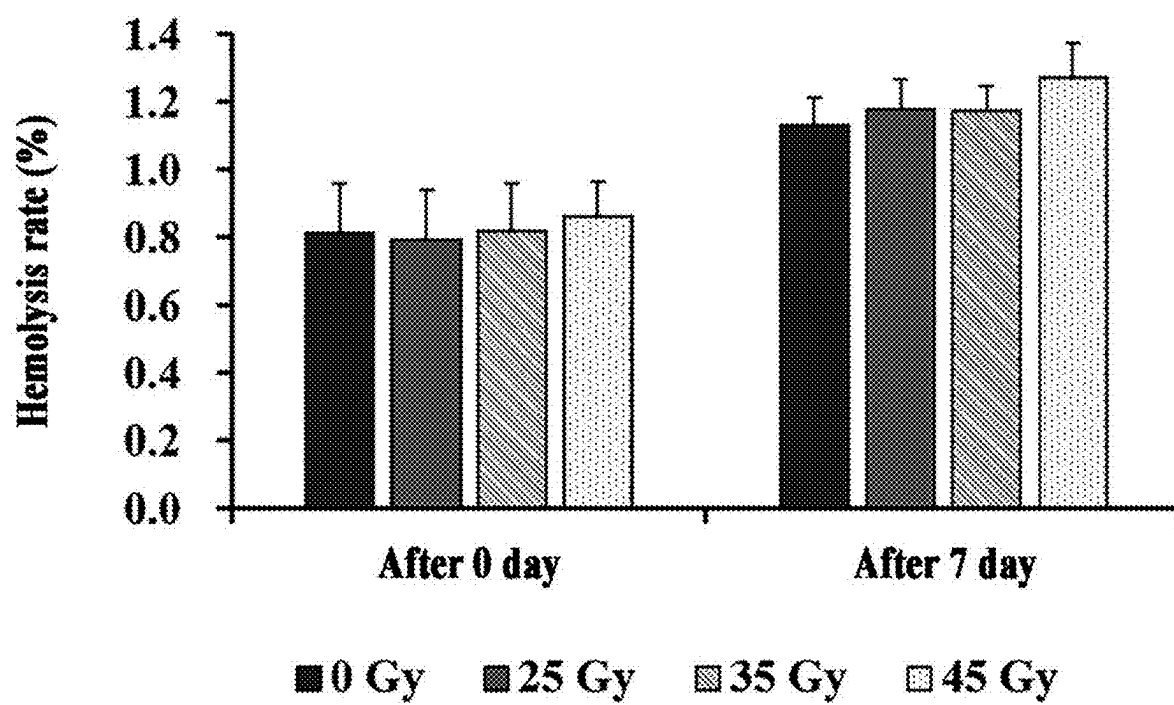


FIG. 2A

Day 0	Exp. 1	Exp. 2	Exp. 3	Exp. 4	Average
0 Gy	0.963	0.994	0.918	0.392	0.817
25 Gy	0.974	1.048	0.752	0.414	0.797
35 Gy	0.947	1.076	0.823	0.432	0.819
45 Gy	1.008	1.023	0.832	0.596	0.865

Day 7	Exp. 1	Exp. 2	Exp. 3	Exp. 4	Average
0 Gy	1.149	1.086	1.349	0.934	1.130
25 Gy	1.226	1.134	1.396	0.965	1.180
35 Gy	1.242	1.171	1.315	0.966	1.174
45 Gy	1.256	1.147	1.561	1.129	1.273

FIG. 2B

Experiment	Dose	Absorbance (nm)			Absorbance _free Hb	Plasma Hb (mg/dL)	Total Hb (mg/dL)	HCT (%)	Hemolysis rate (%)
		415nm	380nm	450nm					
Day 0									
Exp. 1	0 Gy	1.769	0.517	0.539	2.481	207.4	13100	39.2	0.96
	25 Gy	1.785	0.525	0.536	2.510	209.8	13100	39.2	0.97
	35 Gy	1.717	0.490	0.503	2.441	204.0	13100	39.2	0.95
	45 Gy	1.832	0.529	0.535	2.599	217.3	13100	39.2	1.01
Exp. 2	0 Gy	2.087	0.662	0.659	2.853	238.5	13933	41.9	0.99
	25 Gy	2.216	0.724	0.701	3.007	251.3	13933	41.9	1.05
	35 Gy	2.256	0.728	0.696	3.089	258.2	13933	41.9	1.08
	45 Gy	2.117	0.659	0.639	2.936	245.4	13933	41.9	1.02
Exp. 3	0 Gy	2.015	0.611	0.636	2.784	232.7	14400	43.2	0.92
	25 Gy	1.638	0.476	0.519	2.282	190.7	14400	43.2	0.75
	35 Gy	1.765	0.498	0.538	2.495	208.5	14400	43.2	0.82
	45 Gy	1.786	0.506	0.542	2.524	211.0	14400	43.2	0.83
Exp. 4	0 Gy	1.072	0.412	0.454	1.279	106.9	15000	45	0.39
	25 Gy	1.118	0.422	0.464	1.351	112.9	15000	45	0.41
	35 Gy	1.164	0.440	0.480	1.408	117.7	15000	45	0.43
	45 Gy	1.491	0.508	0.529	1.945	162.6	15000	45	0.60
Day 7									
Exp. 1	0 Gy	2.249	0.784	0.751	2.963	247.7	13100	39.2	1.15
	25 Gy	2.488	0.943	0.871	3.161	264.3	13100	39.2	1.23
	35 Gy	2.441	0.872	0.810	3.201	267.6	13100	39.2	1.24
	45 Gy	2.539	0.957	0.883	3.238	270.7	13100	39.2	1.26
Exp. 2	0 Gy	2.428	0.896	0.843	3.118	260.7	13933	41.9	1.09
	25 Gy	2.608	1.025	0.937	3.255	272.1	13933	41.9	1.13
	35 Gy	2.630	0.995	0.902	3.362	281.1	13933	41.9	1.17
	45 Gy	3.027	1.479	1.285	3.291	275.1	13933	41.9	1.15
Exp. 3	0 Gy	2.845	0.820	0.779	4.091	342.0	14400	43.2	1.35
	25 Gy	2.934	0.842	0.792	4.234	354.0	14400	43.2	1.40
	35 Gy	2.781	0.812	0.763	3.987	333.3	14400	43.2	1.31
	45 Gy	3.278	0.943	0.878	4.735	395.8	14400	43.2	1.56
Exp. 4	0 Gy	2.264	0.774	0.708	3.046	254.6	15000	45	0.93
	25 Gy	2.306	0.766	0.699	3.147	263.0	15000	45	0.96
	35 Gy	2.294	0.751	0.685	3.153	263.5	15000	45	0.97
	45 Gy	2.834	1.060	0.923	3.685	308.0	15000	45	1.13

FIG. 2C

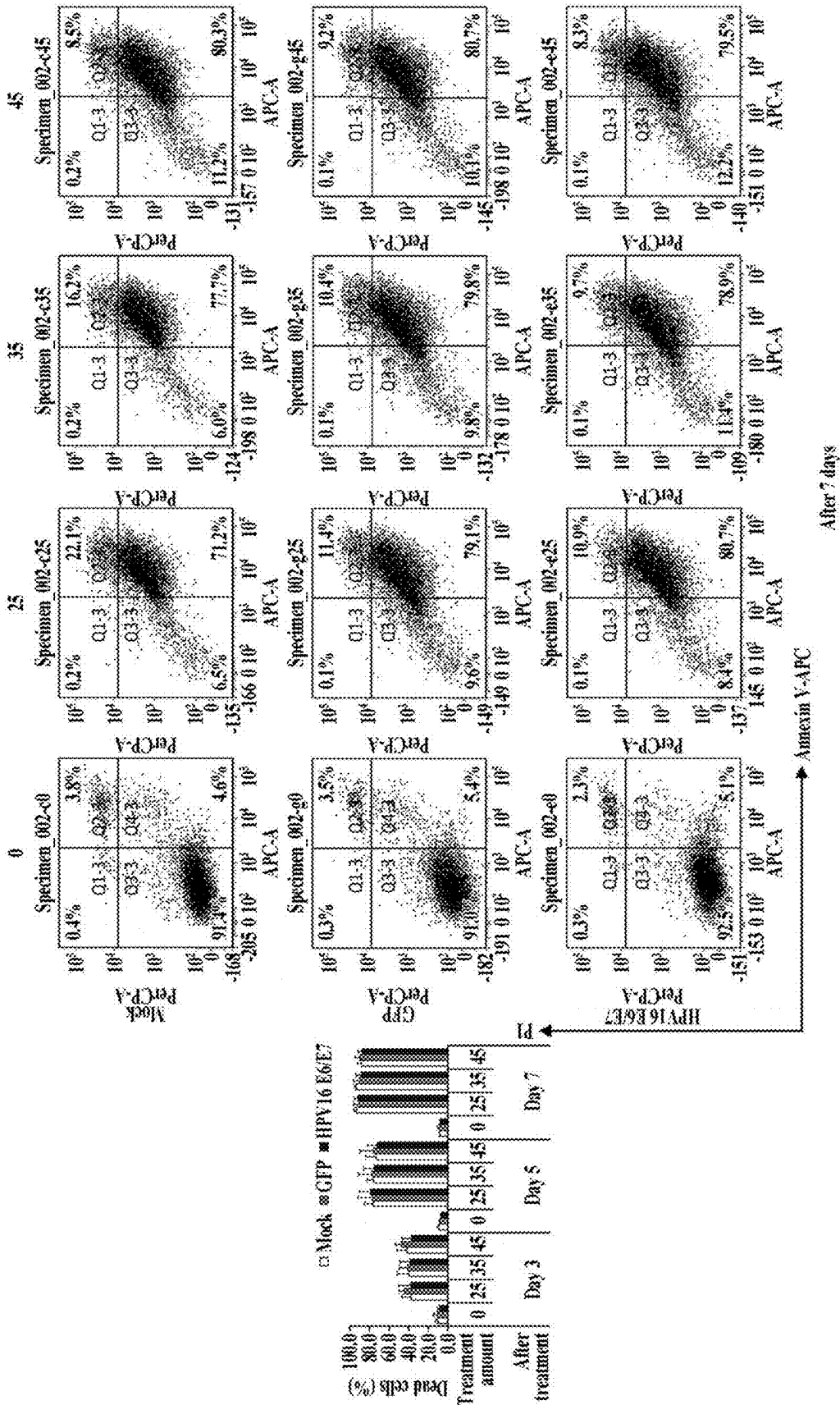


FIG. 3A

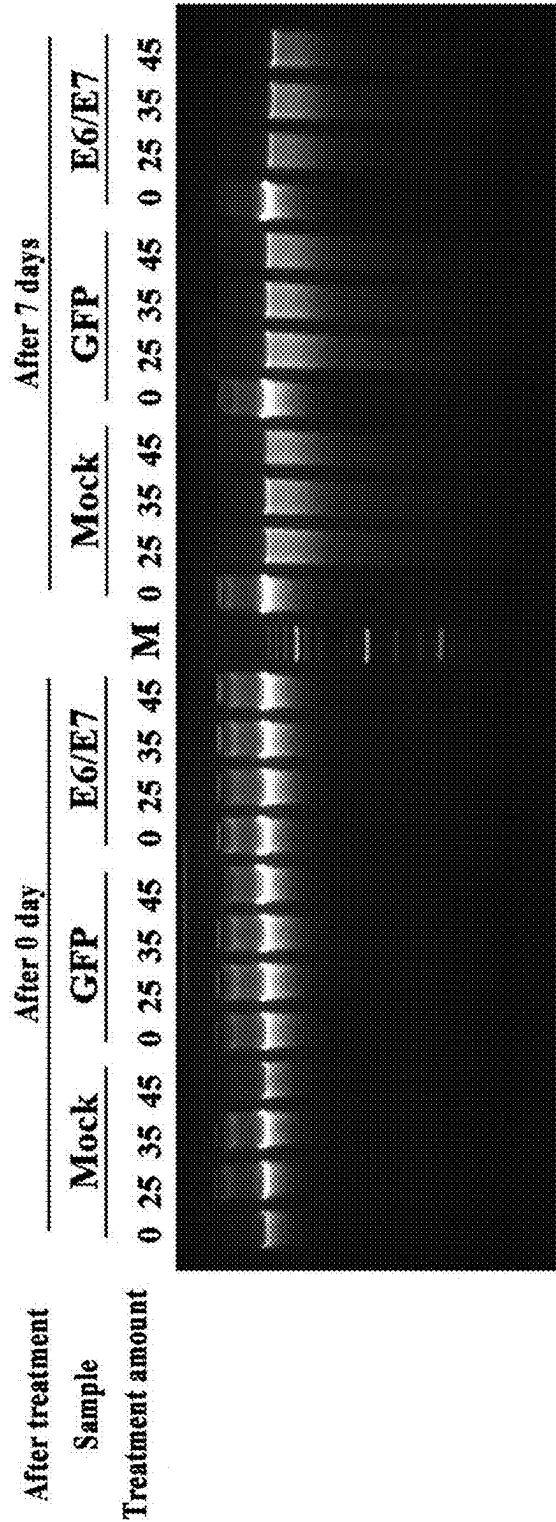


FIG. 3B

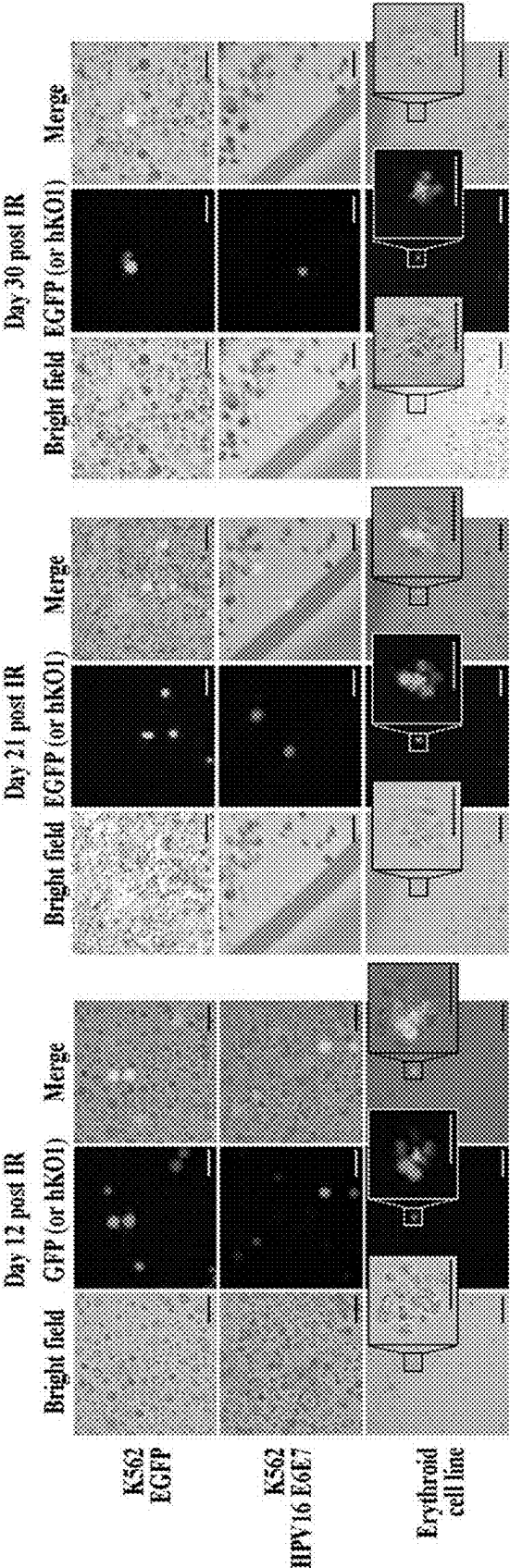


FIG. 3C

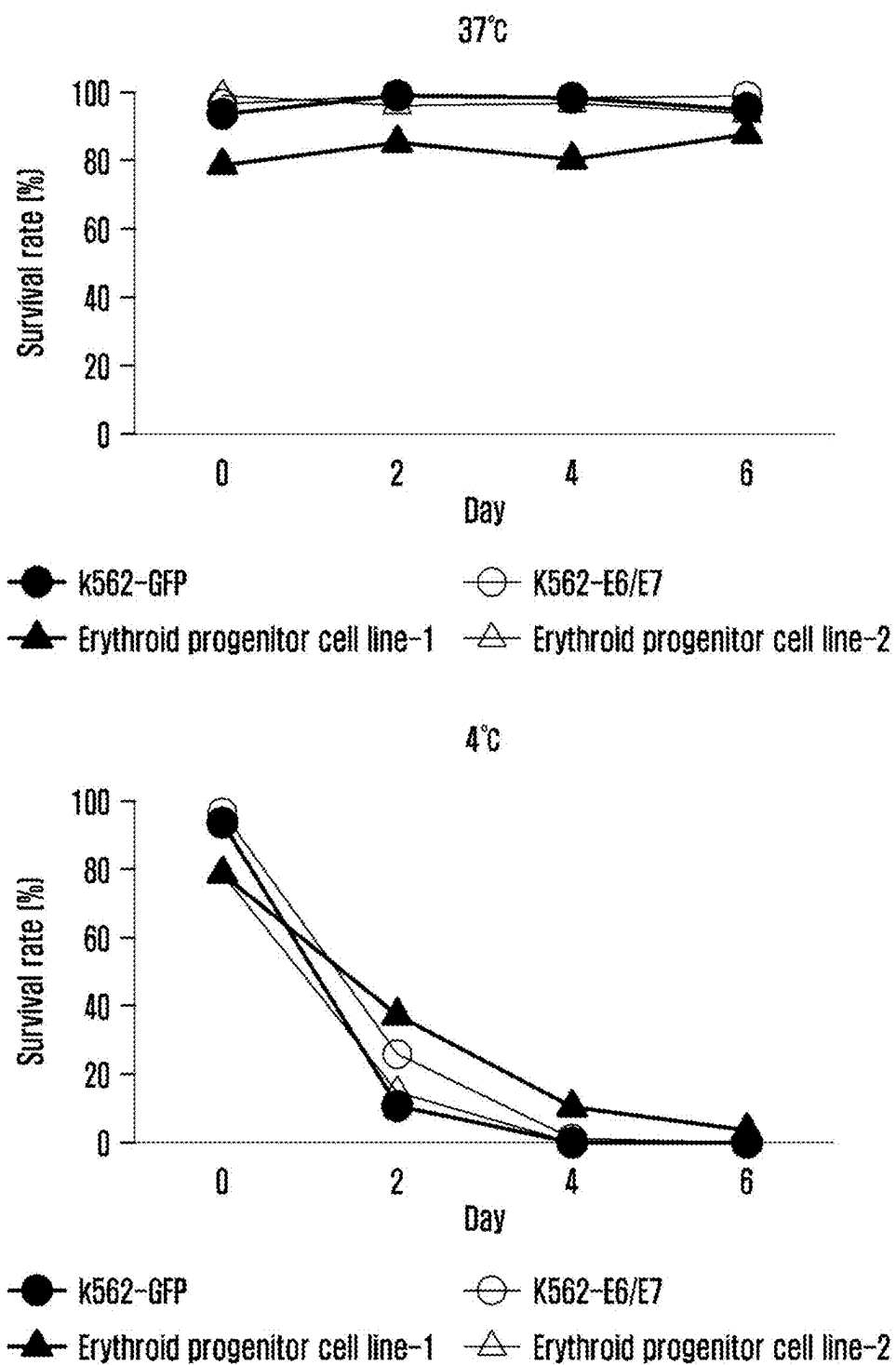


FIG. 4A

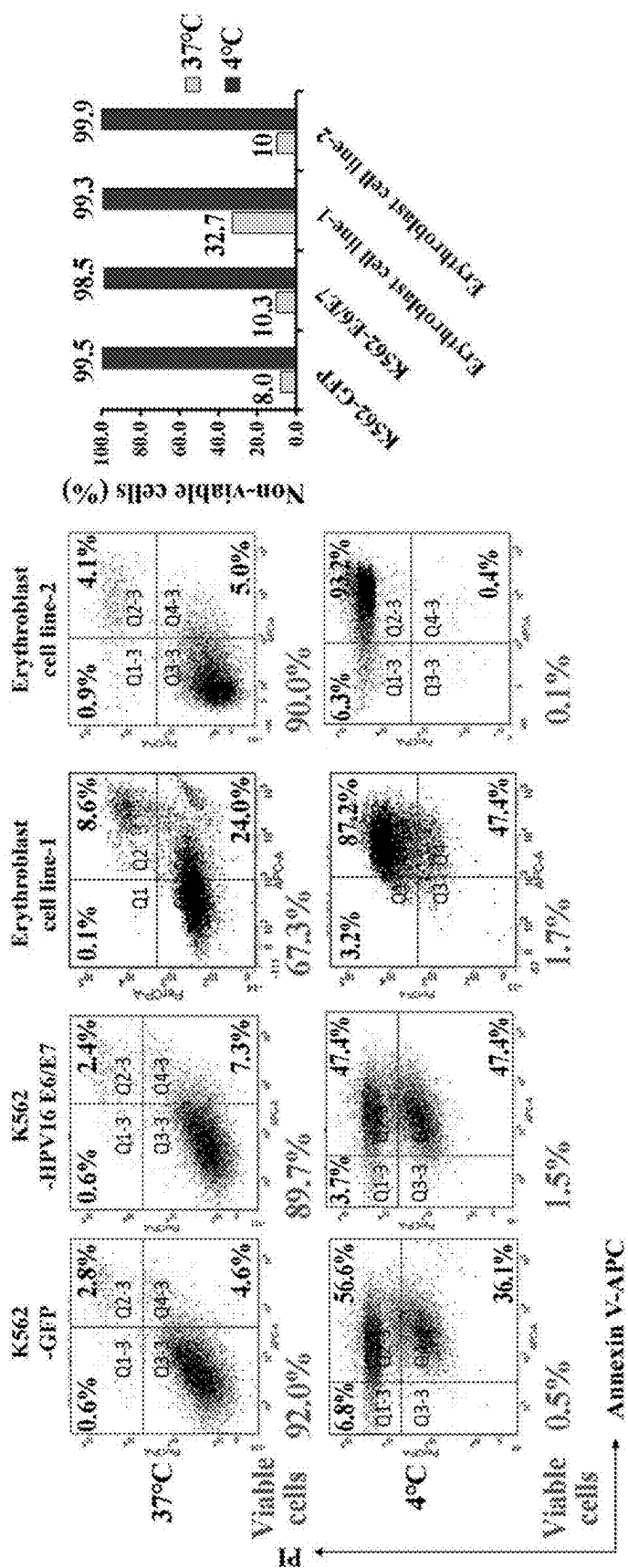


FIG. 4B

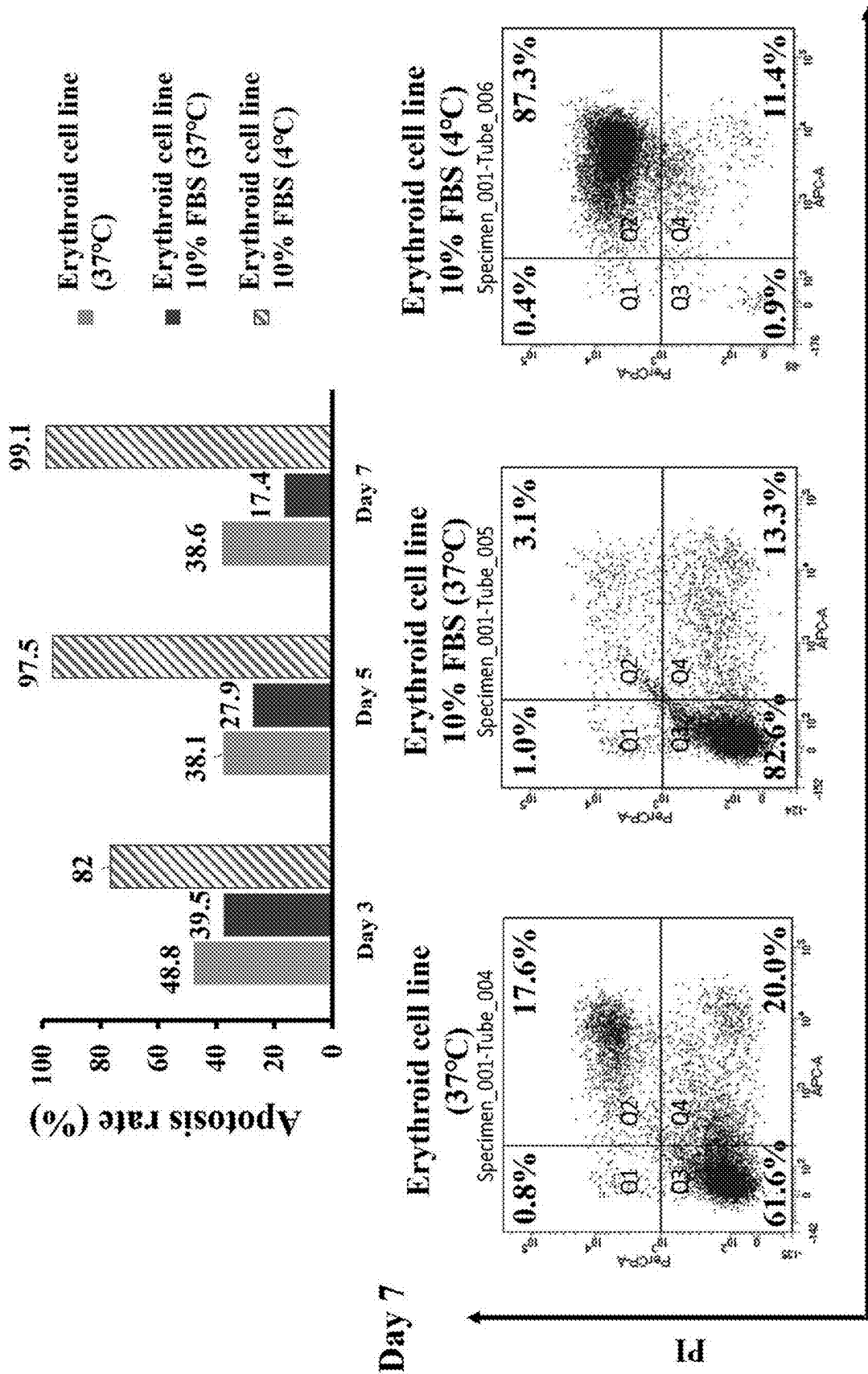


FIG. 4C

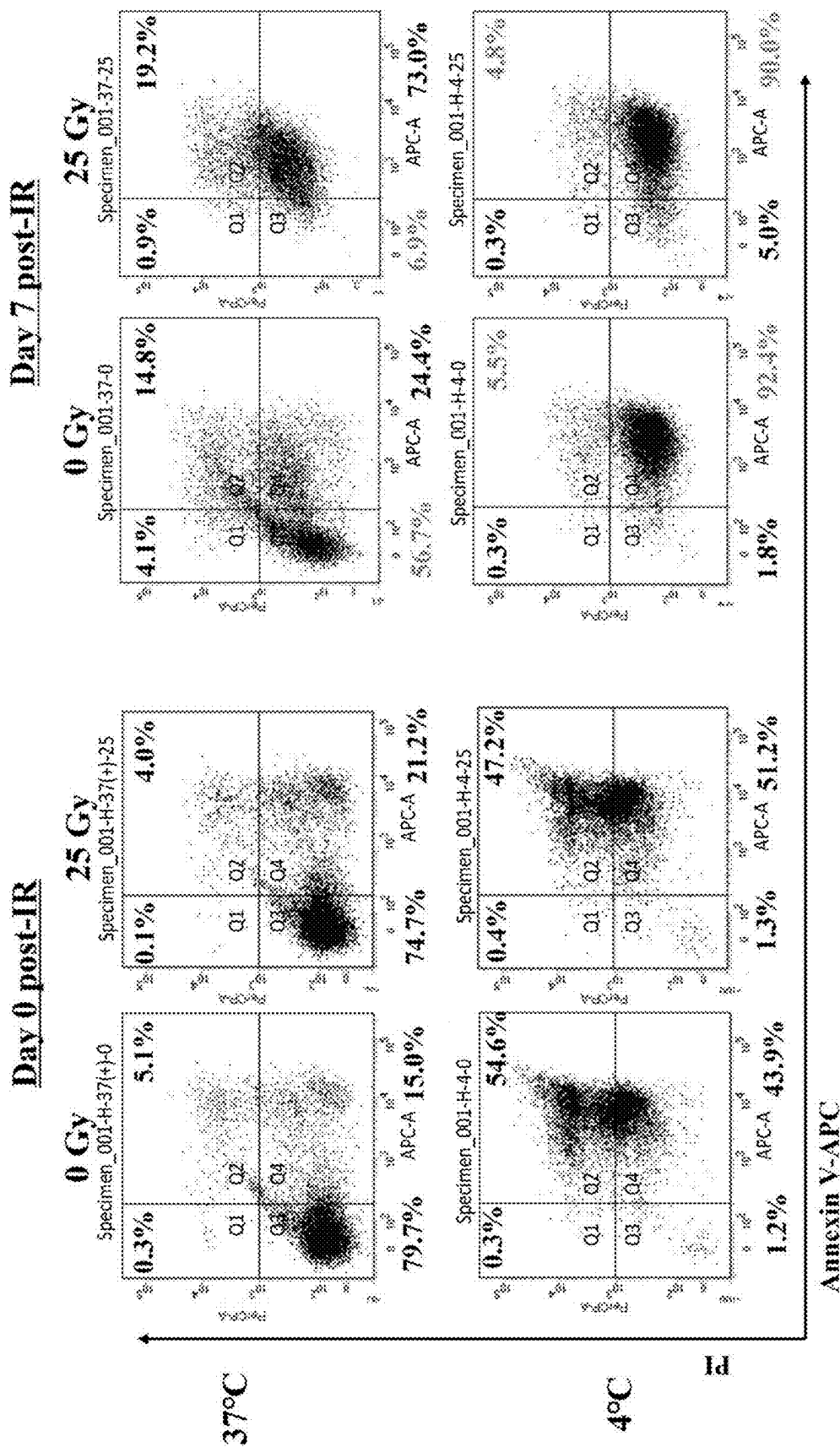


FIG. 5

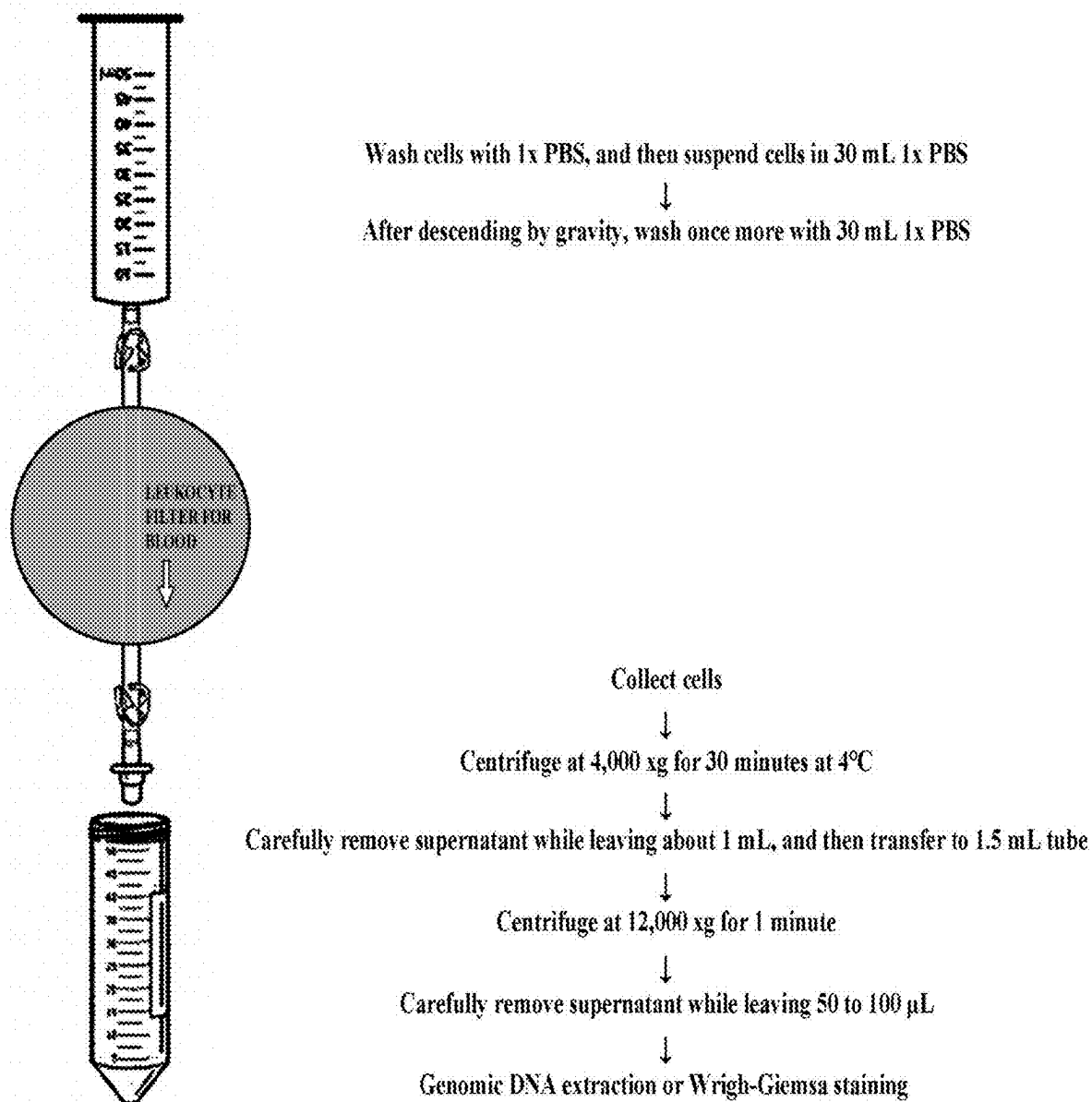


FIG. 6A

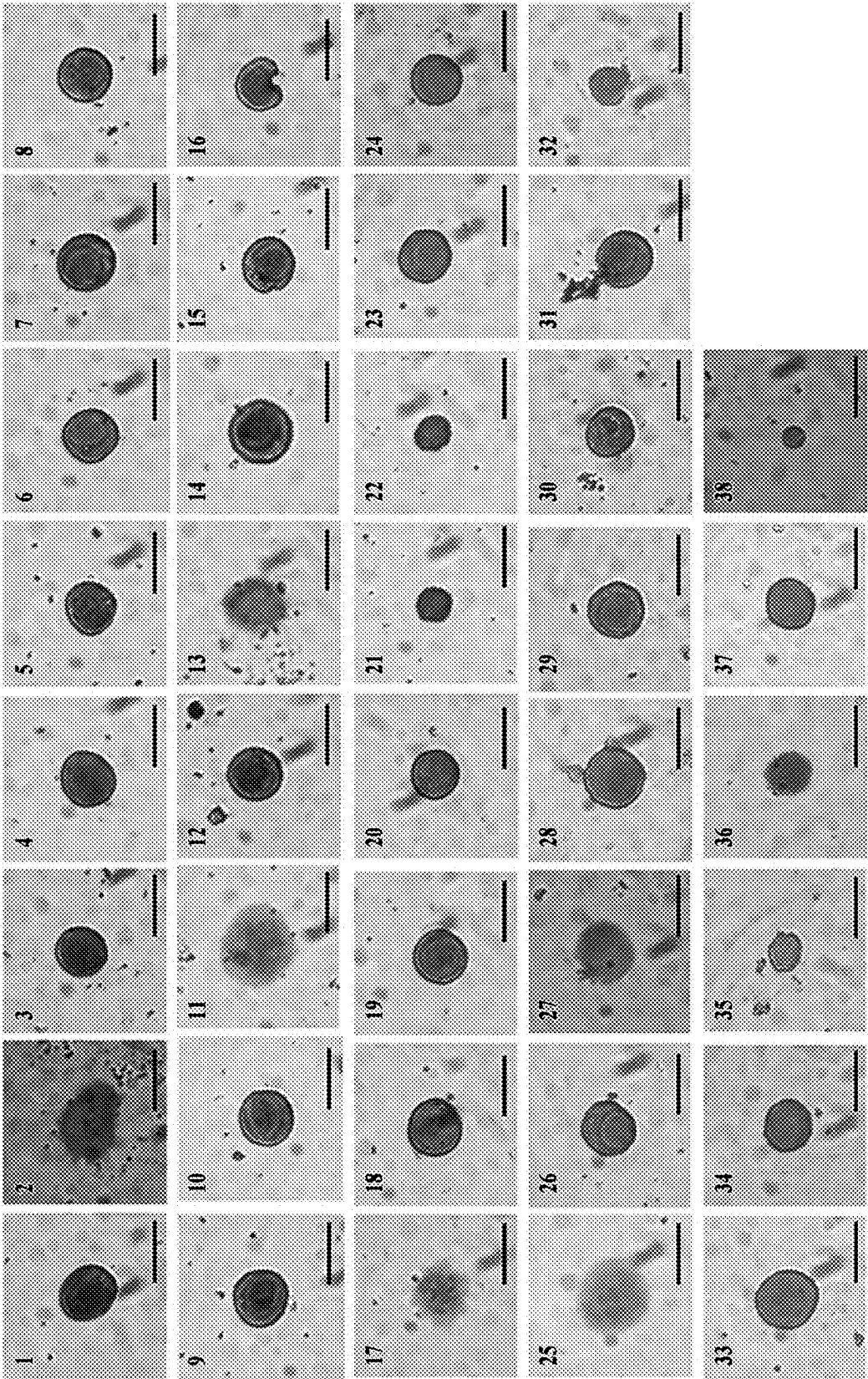


FIG. 6B

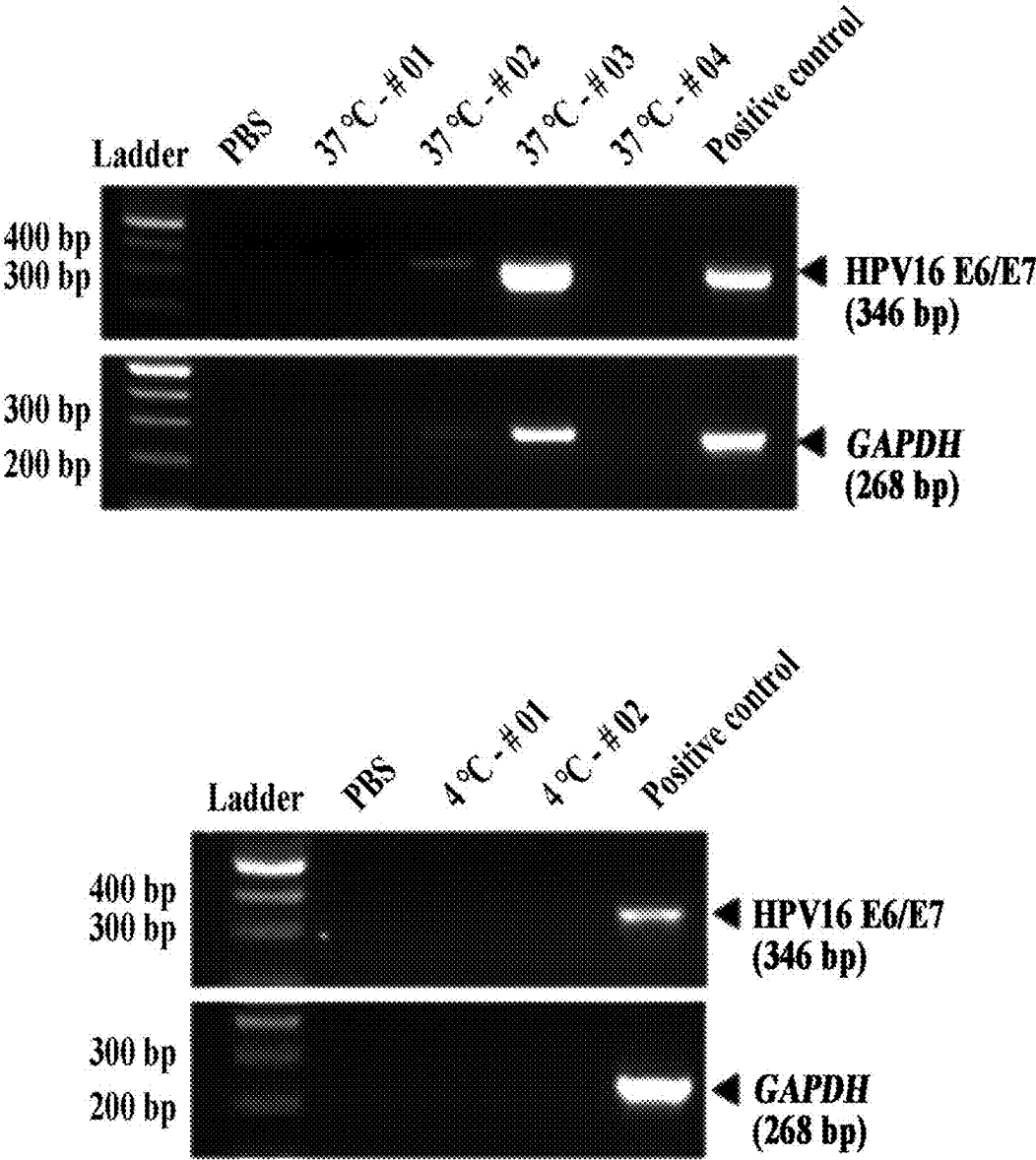


FIG. 6C

NUCLEATED CELL DEATH INDUCTION METHOD USING LOW TEMPERATURE TREATMENT

TECHNICAL FIELD

[0001] The present disclosure relates to a method for killing only nucleated cells and collecting only enucleated blood cells by transferring incubating blood cells to a low temperature.

BACKGROUND ART

[0002] Blood transfusion is a very important treatment for patients who are lacking blood for various reasons. However, blood donated from blood donors has problems such as a shortage of blood supply due to a decrease in the number of blood donations and aging population, the risk of transmission of infectious diseases through blood transfusion, and other side effects of blood transfusion. For this reason, research is being conducted to mass-produce blood cells such as erythrocytes and platelets from stem cells in vitro.

[0003] In particular, in order to mass-produce erythrocytes or platelets, there may be selected a method of differentiating immortal cell lines formed by overexpressing oncogenes in stem cells or erythroid progenitor cells, or induced pluripotent stem cells formed by overexpressing cancer genes. In theory, the erythrocytes produced have no nucleus through an enucleation process, and accordingly, if only the erythrocytes are filtered well and transfused, clinical risks such as causing cancer will not be caused. However, nucleated cells may remain in the final product and a simple, cost-effective and safe method is needed to remove the nucleated cells as much as possible.

[0004] A leukocyte-reducing filter, and the like may be used to collect only the enucleated erythrocytes or platelets. However, this filter does not have 100% of a removal rate of leukocytes, and if there are relatively many nucleated cells among cells before filtering, the filter efficiency decreases and the probability that nucleated cells will not be filtered out in the product after filtering increases.

[0005] In addition, it has been reported in many literatures that blood for transfusion produced in vitro through manipulation such as overexpression of oncogenes may be used safely by killing nucleated cells through irradiation before transfusion, which has not been verified.

[0006] Cell proliferation and survival are affected by various factors, and among the factors, a temperature is an environmental factor with the greatest influence. Enucleated blood cells require relatively fewer metabolites than nucleated cells, which are relatively stable when stored in a refrigerator. Therefore, the present inventors completed the present disclosure by selecting a low temperature treatment method to damage only nucleated cells in a final blood product.

DISCLOSURE OF THE INVENTION

Technical Goals

[0007] An aspect to be achieved by the present disclosure is to provide a nucleated cell death induction method so as to damage only nucleated cells and not to damage enucleated blood cells in incubating blood cells.

Technical Solutions

[0008] In order to solve the aspect, the present inventors propose a method for killing only nucleated cells by a low temperature treatment of incubating blood cells.

[0009] Accordingly, the present disclosure provides a method for inducing the death of nucleated cells in enucleated blood cells

[0010] including incubating hematopoietic stem cells or progenitor cells into enucleated blood cells; and

[0011] killing nucleated cells by a low temperature treatment of the incubating blood cells.

Effects of the Invention

[0012] According to the present disclosure, it is possible to provide a method for inducing the death of nucleated cells by means of a simple, cost-effective and safe method.

BRIEF DESCRIPTION OF DRAWINGS

[0013] FIGS. 1A and 1B show results of confirming that HPV16 E6/E7 is expressed even after Tet off by removing doxycycline.

[0014] FIGS. 2A, 2B and 2C show results of examining the degree of hemolysis of erythrocytes after irradiation.

[0015] FIGS. 3A, 3B and 3C show results of a study on apoptosis according to a radiation dose of nucleated cells.

[0016] FIGS. 4A, 4B and 4C show results of a study on apoptosis according to a low temperature treatment (4° C.) of nucleated cells.

[0017] FIG. 5 shows results of a study on apoptosis according to irradiation and a low temperature treatment.

[0018] FIGS. 6A, 6B and 6C show staining images of erythroid progenitor cells passing through a leukocyte-reducing filter and results of a study of confirming the presence or absence of nucleated cells after passing through a leukocyte-reducing filter after a low temperature treatment (4° C.).

MODE FOR CARRYING OUT THE INVENTION

[0019] Hereinafter, the present disclosure will be described in detail.

[0020] The present disclosure provides a method for inducing the death of nucleated cells in enucleated blood cells,

[0021] including incubating hematopoietic stem cells or progenitor cells into enucleated blood cells; and

[0022] killing nucleated cells by a low temperature treatment of the incubating blood cells.

[0023] In the present disclosure, the “blood cells” refer to platelets, erythrocytes and progenitor cells.

[0024] In the present disclosure, the “hematopoietic stem cells (HSCs)” refer to cells that may potentially differentiate into major components of blood.

[0025] In an embodiment of the present disclosure, the hematopoietic stem cells or progenitor cells may be isolated from bone marrow, umbilical cord blood, or peripheral blood. In an embodiment of the present disclosure, the hematopoietic stem cells or progenitor cells may be derived from all cell source-embryonic stem cells capable of differentiating into erythrocytes or platelets, adult stem cells, erythroid progenitor cell lines, platelet progenitor cell lines, induced pluripotent stem cells (iPS cells), or the like. For example, in the present disclosure, the hematopoietic stem

cells may be isolated from bone marrow, umbilical cord blood, or peripheral blood, or derived from adult stem cells, embryonic stem cells, or iPS cells. In addition, for example, in the present disclosure, the progenitor cells may be isolated from bone marrow, umbilical cord blood or peripheral blood, or derived from adult stem cells, erythroid progenitor cell lines, platelet progenitor cell lines, embryonic stem cells or iPS cells.

[0026] In the present disclosure, the “progenitor cell” is an undifferentiated cell with self-replication and differentiation, but is an ultimately differentiated cell of which a type of finally differentiated cell has already been determined.

[0027] The erythrocytes are formed from hematopoietic stem cells through several differentiation steps, but the nucleus gradually condenses and the cell size decreases through burst forming unit erythroid (BFU-E), colony forming unit erythroid (CFU-E), proerythroblast, basophilic erythroblast, polychromatic erythroblast, and orthochromatic erythroblast. The condensed nucleus is enucleated and then becomes a reticulocyte, and when the remaining RNA and micro-organelles disappear, the reticulocyte has a biconcave shape to become a mature erythrocyte (red blood cell, RBC).

[0028] In the present disclosure, the progenitor cells may be blood cells before enucleation. In addition, the progenitor cells include erythroid progenitor cells or platelet progenitor cells. The erythroid progenitor cells may be burst forming unit erythrocytes (BFU-E), colony forming unit erythrocytes (CFU-E), proerythroblasts, basophilic erythroblasts, polychromatic erythroblasts, orthochromatic erythroblasts or mixtures thereof.

[0029] Therefore, in an embodiment of the present disclosure, the erythroid progenitor cells may be selected from the group consisting of burst forming unit erythrocytes (BFU-E), colony forming unit erythrocytes (CFU-E), proerythroblasts, basophilic erythroblasts, polychromatic erythroblasts, and orthochromatic erythroblasts.

[0030] In another embodiment of the present disclosure, the enucleated blood cells include erythrocytes or platelets. The enucleated erythrocytes may be selected from the group consisting of reticulocytes and erythrocytes.

[0031] The hematopoietic stem cells maintain self-renewal within the body and produce erythrocytes and other blood cells throughout life, but may not form a stem cell niches in the body in vitro, and thus do not maintain self-renewal and differentiate to produce a limited number of blood cells. Accordingly, an immortalized erythroid progenitor cell line may be used to induce hematopoietic stem cells to differentiate into blood cells, but increase the production of blood cells while extending the cell proliferation period through regulation of genes related to cell proliferation. However, in this case, since overexpression of oncogenes was used, there is a problem in safety if the final product may not be transfused by completely removing the nucleus.

[0032] Accordingly, an embodiment of the present disclosure includes killing nucleated cells by a low temperature treatment of the incubating blood cells at 0° C. or higher and 15° C. or lower. The incubating blood cells may be understood as a concept including both enucleated cells and non-enucleated cells in a continuous process of incubating hematopoietic stem cells or progenitor cells into enucleated blood cells.

[0033] In an embodiment of the present disclosure, the nucleated cells may be erythroid progenitor cells or platelet progenitor cells that overexpress HPV16 E6/E7. HPV16 E6/E7 refers to a HPV virus gene E6/E7, which is a cancer-causing gene that is not expressed in humans. Desirably, the overexpressed gene required for constructing an erythroid progenitor cell line may be HPV16 E6/E7, but is not limited thereto.

[0034] The present inventors confirmed through experiments that when the incubating blood cells are subjected to a low temperature treatment, non-enucleated nucleated cells are killed. In an embodiment of the present disclosure, the low temperature treatment step may be performed at a temperature of 0° C. or higher and 15° C. or lower. Desirably, the low temperature treatment step may be performed at 0° C. or higher and 10° C. or lower, more desirably 0° C. or higher and 7° C. or lower, or 0° C. or higher and 5° C. or lower, and most desirably 3° C. or higher and 5° C. or lower. The low temperature treatment step may be performed by refrigerating the incubating cells. In an embodiment of the present disclosure, the low temperature treatment step may be performed by refrigeration for 1 to 30 days. The low temperature treatment step may be performed by refrigeration for desirably 3 to 10 days, or 5 to 10 days, and most desirably 7 to 10 days.

[0035] According to the present disclosure, when the incubating blood cells are subjected to the low temperature treatment, the death rate of nucleated cells is measured to be 97% or higher. Desirably, during the low temperature treatment, the death rate of nucleated cells is measured to be 98% or higher, more desirably 99% or higher, and most desirably 99.9% or higher. The cell death rate after the low temperature treatment may be measured using any method known in the art. For example, the measurement of the cell death rate may be measured by a flow cytometer, but is not limited thereto.

[0036] In an embodiment of the present disclosure, the method may further include a step of passing the incubating blood cells through a leukocyte-reducing filter to kill only nucleated cells in the incubating blood cells. The step is desirably performed after the low temperature treatment step, but is not limited thereto.

[0037] In yet another embodiment of the present disclosure, the method may further include a step of irradiating the incubating blood cells to kill only the nucleated cells in the incubating blood cells. The irradiating step may be performed before the low temperature treatment step or may also be performed after the low temperature treatment step. The radiation dose may be used in various intensities. For example, the total radiation dose to be administered may be about 10 Gy (gray) to about 60 Gy, about 10 Gy to about 50 Gy, about 20 Gy to about 50 Gy, about 20 Gy to about 45 Gy, about 20 Gy to about 35 Gy, about 20 Gy to about 30 Gy, or about 25 Gy to about 30 Gy, but is not limited thereto.

[0038] In an embodiment of the present disclosure, the method may further include a step of treating the incubating blood cells with nuclease. In the present disclosure, the term “nuclease” is a general term for nucleic acid decomposing enzymes, and may vary depending on a use and a location. In general, the nuclease may cause a reaction that breaks diphosphorylation bonds of a nucleic acid molecule such as DNA or RNA, and examples of the nuclease include endonuclease that cleaves the inside of a nucleic acid and exonuclease that cleaves the end of the nucleic acid. The

nuclease additionally treated in the present disclosure may cleave nucleic acids released from the nuclei or cells present in a culture medium, and the like.

[0039] According to the present disclosure, the enucleated blood cells collected from the low temperature-treated cells may not be substantially hemolyzed. In an embodiment of the present disclosure, the enucleated blood cells may be enucleated erythrocytes, and the hemolysis rate (%) of the enucleated erythrocytes may be less than 5, desirably less than 3, less than 1, and more desirably less than 0.8.

[0040] Hereinafter, preferred Examples and Experimental Examples will be presented in order to assist the understanding of the present disclosure. However, the following Examples and Experimental Examples are just provided to more easily understand the present disclosure, and the contents of the present disclosure are not limited by Examples or Experimental Examples.

EXAMPLE

Experimental Method

1) Sample Collection

[0041] For fresh erythrocyte samples, an experiment was conducted on the residual blood of patients aged 20 to 60 years who had completed tests and whose erythrocyte count was within a normal range (hemoglobin 12.0 g/dL or higher) among specimens for general blood tests after obtaining consent. For collection of umbilical cord blood, the consent was obtained from healthy mothers, and then the umbilical cord blood was collected after birth to isolate CD34+ cells. In addition, K562 erythroleukemia cells or K562 cells overexpressing HPV16 E6/E7, and erythroid progenitor cell lines were used.

2) Cell Production and Incubation

[0042] A lentiviral tetracycline-regulated system (lentiviral Tet-on system) capable of regulating gene expression with tetracycline was used to construct erythroid progenitor cell lines overexpressing HPV16 E6/E7. When inducing erythrocyte differentiation from the erythroid progenitor cell lines, antibiotics were removed and it was shown using PCR that HPV16 E6/E7 mRNA was not expressed.

[0043] The K562 cells were incubated in a RPMI 1640 medium supplemented with 10% fetal bovine serum (FBS), and the medium was replaced with a fresh medium supplemented with doxycycline (1 µg/mL) every 2 to 3 days.

[0044] The erythroid progenitor cell line was incubated in a Stemspan SFEM medium supplemented with EPO (3 IU/ml), SCF (50 ng/ml), and dexamethasone (1 µM). To provide an environment in which the cell line grew well, some conditions were replaced with a fresh medium at a concentration of 2.5×10⁵ cells/ml by adding 10% FBS every 2 to 3 days.

[0045] The FBS was used with Tet system-approved FBS without containing a tetracycline-derived substance to exclude a background that may affect the tetracycline-induced system.

3) Irradiation

[0046] A gamma irradiation device Biobeam8000 (STS Steuerungstechnik &. Strahlenschutz GmbH, Braunschweig, Germany, Cs-137) was used to irradiate at radia-

tion doses of 0 Gy, 25 Gy, 35 Gy, and 45 Gy (radiation dose of approximately 1.4 Gy/min).

4) Measurement of Hemolysis Rate of Erythrocytes

[0047] Blood was placed in a 1.5 ml tube and centrifuged at 2000×g for 10 minutes at 10° C., and 100 µl of the supernatant (plasma layer) was placed in a 96-well plate. After measuring the absorbance at 415 nm, 380 nm, and 450 nm using a spectrophotometer, plasma hemoglobin (Hb) (mg/dL) was calculated using the following Equation.

Plasma Hb(mg/dL) = 83.6 × (2 × A415 − A380 − A450)

[0048] In addition, the hemolysis rate of erythrocytes was calculated according to Equation below. (Hct=hematocrit)

Hemolysis rate of erythrocytes (%) = $\frac{\text{Plasma Hb} \times (100 - \% \text{ Hct})}{\text{Total Hb}} \times 100$

5) Study on Apoptosis Using Flow Cytometry

[0049] A flow cytometer Canto II (BD Biosciences, San Jose, CA) was used to measure apoptosis after irradiation and 4° C. low temperature treatment. The cells were harvested, washed with phosphate-buffered saline (PBS), suspended in a 1× binding buffer, and the sample was added with each 5 µl of annexin V and propidium iodide (PI; BD Biosciences) and reacted for 15 minutes at room temperature under light blocking, and then the apoptosis was measured using a flow cytometer.

6) Measurement of DNA Fragmentation

[0050] The cells were harvested and washed with PBS, and then genomic DNA was extracted using the PureLink genomic DNA mini kit (Invitrogen). 200 ng of purified genomic DNA was loaded on a 1% agarose gel and electrophoresis was performed at 50 V of voltage for 75 minutes.

7) PCR

[0051] Pre-denaturation was performed using Amfisure PCR master mix (GenDEPOT) at 96° C. for 30 seconds, and then 35 cycles of denaturation at 96° C. for 15 seconds, annealing at 65° C. for 30 seconds, and extension at 72° C. for 20 seconds were performed. The primers used were as follows.

TABLE 1	
HPV16 E6/E7	
Forward (SEQ ID NO: 1)	5'-GTCAAAGCCACTGTGTCCTG-3'
Reverse (SEQ ID NO: 2)	5'-GATTTGCAACCAGAGACAACTG-3'

TABLE 2	
GAPDH	
Forward (SEQ ID NO: 3)	5'-GAAGGTGAAGGTCGGAGTC-3'
Reverse (SEQ ID NO: 4)	5'-GACAAGCTTCCCGTTCTCAG-3'

Experimental Example 1. Confirmation of
Expression of Overexpressed HPV16 E6/E7 after
Removal of Doxycycline

[0052] To confirm whether the expression of oncogene HPV16 E6/E7 was well regulated by a Tet-on system regulated by the presence or absence of doxycycline, an experiment was conducted to remove doxycycline from K562 cells overexpressing HPV16 E6/E7. It was confirmed by cell fluorescence imaging and PCR technique that gene expression was not completely turned off due to a leakage effect, in which the background of the vector was expressed even in the absence of doxycycline.

[0053] An Enhanced Green Fluorescent Protein (eGFP) sequence was designed to be expressed upstream of HPV16 E6/E7, and when the doxycycline was present, a transactivator bound to a TRE promoter to regulate the expression.

[0054] The cells were washed twice with PBS, and then the medium was replaced with a doxycycline-free medium, and the fluorescence expression levels after 3 and 6 days were compared with those when the cells were grown in the doxycycline-free medium. It was observed under a fluorescence microscope that eGFP fluorescence was not completely eliminated (FIG. 1A). FIG. 1A is a cell fluorescence photograph (Scale bar=200 μm) taken at 0, 3, and 6 days after doxycycline was removed from a K562 cell culture overexpressing eGFP-HPV16 E6/E7.

[0055] To confirm the mRNA level of HPV16 E6/E7 after doxycycline removal, the cells were harvested at 3 and 6 days after doxycycline removal, RNA was extracted, and cDNA was synthesized. After PCR gene amplification, the presence of HPV16 E6/E7 mRNA was shown using electrophoresis (FIG. 1B). FIG. 1B shows the mRNA expression of HPV16 E6/E7 confirmed by post-PCR electrophoresis after 3 and 6 days of incubation depending on the presence or absence of doxycycline, and Glyceraldehyde 3-phosphate dehydrogenase (GAPDH) was used as an internal control.

[0056] As a result, although the tetracycline-induced expression system selectively suppressed oncogene expression, as a result of observation by fluorescence microscopy and post-PCR electrophoresis, it was confirmed that the Tet-on system, in which the HPV16 E6/E7 gene was expressed even after doxycycline was removed, was not perfect, and there was a problem with the continued expression of introduced genes (background expression). That is, although erythrocytes as the final product have no nucleus due to the enucleation process, it may be confirmed that a method for killing nucleated cells remaining in the final erythrocyte product is required to secure stability suitable for clinical application.

Experimental Example 2. Test of Hemolyzed
Erythrocyte Ratio after Irradiation

[0057] To determine the optimal radiation absorbed dose that may damage nucleated cells while minimizing hemolysis

of erythrocytes, the hemolysis rate of erythrocytes at 0 day after irradiation and 7 days after refrigeration (4° C.) was calculated using the following Equation.

$$\text{Hemolysis (\%)} = \frac{\text{Plasma Hb} \times (100 - \% \text{ Hct})}{\text{Total Hb}}$$

$$\text{Plasma Hb (mg/dl)} = 83.6 \times (2 \times A_{415} - A_{380} - A_{450})$$

Hb: Plasma hemoglobin (Hemoglobin)

A415: Absorbance at 415 nm wavelength

A380: Absorbance at 380 nm wavelength

A450: Absorbance at 450 nm wavelength

Hct: Hemolysis rate of erythrocytes

(Hematocrit, ratio of erythrocytes in blood)

[0058] The hemolysis rates of erythrocytes were compared at 25 Gy, 35 Gy, and 45 Gy relative to 0 Gy of radiation dose after 7 days of irradiation, and there was no significant difference in hemolysis rate after 7 days of refrigeration according to a radiation dose (n=4, 0 Gy vs 25 Gy: p=0.699, 0 Gy vs 35 Gy: p=0.714, 0 Gy vs 45 Gy: p=0.318 Exp.: experiment). The hemolysis rates of erythrocytes immediately after irradiation were average 0.817, 0.797, 0.819, and 0.865 for 0 Gy, 25 Gy, 35 Gy, and 45 Gy, respectively, and after 7 days of refrigeration (4° C.), the hemolysis rates were 1.130, 1.180, 1.174, and 1.273, respectively (FIGS. 2A and 2B).

[0059] The absorbance measured to calculate the hemolysis rate of erythrocytes and the hemolysis rate of erythrocytes after irradiation were shown in FIG. 2C. The evaluation criteria for the hemolysis rate of erythrocytes are less than 1% based on the U.S. Food and Drug Administration (FDA) and less than 0.8% based on the European Council of Europe (CE).

Experimental Example 3. Study on Apoptosis
According to Radiation Dose of Nucleated Cells

[0060] To ensure stability suitable for clinical application, it is necessary to remove nucleated cells remaining in the final erythrocyte product, and as an experiment for verifying the removal, K562 cells overexpressing oncogene HPV16 E6/E7 were irradiated at 0, 25, 35, and 45 Gy of radiation.

[0061] After irradiation, cells were incubated at 37° C., and flow cytometry was performed using Annexin V/PI staining after 3, 5, and 7 days of irradiation to measure apoptosis. As a result, 7 days after irradiation, a mock (a control group in which only a reagent was added without transduction), an eGFP control group, and a HPV16 E6/E7 overexpression sample all showed approximately 90% of apoptosis, and there was no significant difference between the samples (FIG. 3A). FIG. 3A shows results of flow cytometry after annexin v/PI staining at 3, 5, and 7 days after irradiation of 0, 25, 35, and 45 Gy of radiation to mock cells without virus treatment, eGFP control, or K562 cells overexpressing HPV16 E6/E7. In addition, after extracting genomic DNA from the sample, the fragmented DNA was confirmed through electrophoresis (FIG. 3B). FIG. 3B shows results of electrophoresis performed at 50 V of voltage for 75 minutes by loading 200 ng of purified

genomic DNA on a 1% agarose gel after extracting the genomic DNA from the sample of FIG. 3A.

[0062] Even by increasing the radiation dose from 25 Gy to 35 and 45 Gy, there was no significant increase in the apoptosis rate. As a result, it may be confirmed that complete apoptosis of nucleated cells is not induced only by irradiation. In addition, the radiation dose widely used for blood products in domestic medical institutions was 25 Gy (The Korean Journal of Blood Transfusion, Vol. 27, No. 2, Current Status of Irradiated Blood Components and Blood Irradiators in Korean Medical Institutions, 2016), and thus the radiation dose was set to 25 Gy in subsequent experiments.

[0063] K562 cells overexpressing GFP or HPV16 E6/E7 and an erythroid progenitor cell line overexpressing HPV16 E6/E7 were irradiated with 25 Gy of radiation, and then incubated at 37° C. and the expression of fluorescence was observed, and cells expressing fluorescence could be observed up to 30 days in incubation (FIG. 3C). It is shown that irradiation alone is not sufficient for nucleated cell death, as safety issues may occur when applied clinically.

Experimental Example 4. Study on Apoptosis According to Low Temperature Treatment (4° C.) of Nucleated Cells

[0064] As confirmed in Experimental Example 3, since irradiation did not induce complete apoptosis, to confirm whether complete apoptosis may be induced by refrigeration at 4° C., the degree of apoptosis was compared using trypan blue staining after refrigeration at 4° C. in K562 cells overexpressing HPV16 E6/E7 and an erythroid progenitor cell line.

[0065] During incubation at 37° C., a cell survival rate of 75% or higher was shown, but during refrigeration at 4° C., the cell survival rate was decreased immediately, and on day 6 of storage, K562 and erythroid progenitor cell line-2 showed the cell survival rate of less than 1%, and erythroid progenitor cell line-1 showed the cell survival rate of about 3% (trypan blue staining). After day 6 of refrigeration at 4° C., as a result of measuring the apoptosis rate by flow cytometry using annexin V/PI staining, an apoptosis rate of 98.5% or higher was shown (FIGS. 4A and 4B). FIG. 4A illustrates a result of measuring the cell survival rate when nucleated cells were stored in a refrigerator at 4° C., in which when the K562 cells, the K562 cells overexpressing HPV16 E6/E7 and the erythroid progenitor cell lines-1 and -2, were incubated at 37° C. and stored in a refrigerator at 4° C., the cell survival rates were measured by staining with trypan blue. FIG. 4B illustrates results of measuring the cell survival rate through flow cytometry after staining cells with Annexin V/PI staining at 6 days after refrigeration at 4° C.

[0066] When maintaining and incubating the erythroid progenitor cell lines, serum was not added, and then the medium was added with the serum when differentiating into erythrocytes, and thus apoptosis was measured by flow cytometry after annexin V/PI staining after 3, 5, and 7 days according to the conditions. The refrigeration was per-

formed by adding 10% fetal bovine serum (FBS) to the medium, similarly to the storage conditions for erythrocytes (FIG. 4C). FIG. 4C illustrates results of measuring the cell survival rates of the erythroid progenitor cell line overexpressing HPV16 E6/E7 after 7 days of incubation at 37° C. (serum +/-) and refrigeration at 4° C. (serum +).

[0067] As a result, the apoptosis was observed as 82% after 3 days, 97.5% after 5 days, and 99.1% after 7 days of 4° C. refrigeration.

Experimental Example 5. Comparison of Apoptosis According to Irradiation And Low Temperature Treatment

[0068] In order to confirm induction of apoptosis by a combination of irradiation (25 Gy) and refrigeration (4° C.) in addition to irradiation (25 Gy) or refrigeration (4° C.) alone, after culturing and storing for 4 days at 37° C. and 4° C. in a medium supplemented with 10% serum, cells were treated with irradiation (25 Gy) and Annexin V/PI staining was performed after 4 hours (IR-Day 0) and 7 days (IR-Day 7), and apoptosis was measured using flow cytometry. As a result, the apoptosis rate (98.2%) in a non-irradiated condition was higher than the apoptosis rate (95.1%) in the case of combined treatment with irradiation (25 Gy) and refrigeration (FIG. 5).

Experimental Example 6. Experiment on Removal of Erythroid Progenitor Cells Using Leukocyte-Reducing Filter

[0069] To determine how much nucleated cells may be removed using a leukocyte-reducing filter, 5.58×10^7 cells of erythroid progenitor cells passed through a BioR 02 plus BS PF (Fresenius kabi, A2BB0080) leukocyte-reducing filter and contained in a 50-ml conical tube (FIG. 6A). After passing through the leukocyte-reducing filter and centrifuging for 20 minutes, the collected cells were stained with Wright-Giemsa, and a total of 38 cells were observed (Scale bar=20 μ m) (FIG. 6B).

[0070] Since 99.9% or more of cells may be removed, but some passing cells are present by using the leukocyte-reducing filter, nucleated cells may not be completely removed only by the leukocyte-reducing filter, and the removal rate of erythroid progenitor cells may be increased by using a method of using the leukocyte-reducing filter in a state where most cells die by refrigeration at 4° C.

[0071] The erythroid progenitor cell lines incubated at 37° C. and 4° C. for 7 days each passed through the leukocyte-reducing filter, and then genomic DNA was extracted, amplified by PCR, and confirmed by electrophoresis to determine whether there were any nucleated cells remaining in a filtrate. As a result of the experiment repeated four times under the condition of 37° C. and repeated twice under the condition of 4° C., it was verified that there were no nucleated cells because genomic DNA was not amplified when passing through the leukocyte-reducing filter after incubation at 4° C. (FIG. 6C).

SEQUENCE LISTING

Sequence total quantity: 4
SEQ ID NO: 1 moltype = DNA length = 21

-continued

FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 1		
gtcaaaagcc actgtgtcct g		21
SEQ ID NO: 2	moltype = DNA length = 22	
FEATURE	Location/Qualifiers	
source	1..22	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 2		
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SEQ ID NO: 3	moltype = DNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 3		
gaaggtgaag gtcggagtc		19
SEQ ID NO: 4	moltype = DNA length = 20	
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	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 4		
gacaagcttc ccgttctcag		20

1. A method for nucleated cell death induction in incubating blood cells, comprising:

incubating hematopoietic stem cells or progenitor cells into enucleated blood cells; and

killing nucleated cells by a low temperature treatment of the incubating blood cells, wherein the low temperature treatment is performed at 0° C. or higher and 15° C. or lower.

2. The method for nucleated cell death induction of claim 1, wherein the hematopoietic stem cells or progenitor cells are isolated from bone marrow, umbilical cord blood, or peripheral blood.

3. The method for nucleated cell death induction of claim 1, wherein the progenitor cells comprise erythroid progenitor cells.

4. The method for nucleated cell death induction of claim 1, wherein the enucleated blood cells comprise enucleated erythrocytes or enucleated platelet cells.

5. The method for nucleated cell death induction of claim 4, wherein the enucleated erythrocytes are selected from the group consisting of reticulocytes and erythrocytes.

6. The method for nucleated cell death induction of claim 1, wherein the nucleated cells are erythroid progenitor cells or platelet progenitor cells that overexpress proliferation-related genes.

7. The method for nucleated cell death induction of claim 6, wherein the proliferation-related gene is HPV16 E6/E7.

8. The method for nucleated cell death induction of claim 1, wherein the low temperature treatment is performed by refrigeration for 3 to 10 days.

9. (canceled)

10. The method for nucleated cell death induction of claim 1, wherein the low temperature treatment is performed at 0° C. or higher and 5° C. or lower.

11. The method for nucleated cell death induction of claim 1, further comprising:

passing the incubating blood cells through a leukocyte-reducing filter.

12. The method for nucleated cell death induction of claim 1, further comprising:

irradiating the incubating blood cells.

13. The method for nucleated cell death induction of claim 1, further comprising:

treating nuclease.

14. The method for nucleated cell death induction of claim 1, wherein the death rate of nucleated cells in the low temperature treated cells is 97% or higher.

15. The method for nucleated cell death induction of claim 1, wherein the hematopoietic stem cells or progenitor cells are derived from adult stem cells, erythroid progenitor cell lines, platelet progenitor cell lines, embryonic stem cells, or iPS cells.

16. The method for nucleated cell death induction of claim 3, wherein the erythroid progenitor cells are selected from the group consisting of burst forming unit erythroids (BFU-E), colony forming unit erythroids (CFU-E), pro-erythroblasts, basophilic erythroblasts, polychromatic erythroblasts, and orthochromatic erythroblasts.

17. The method for nucleated cell death induction of claim 1, wherein the low temperature treatment is performed by refrigeration for 3 to 10 days, and

wherein the low temperature treatment is performed at 0° C. or higher and 5° C. or lower.

18. The method for nucleated cell death induction of claim 1, further comprising:

passing the incubating blood cells through a leukocyte-reducing filter;

irradiating the incubating blood cells; and

treating nuclease.

19. A method for producing red blood cell, comprising: differentiating erythroid progenitor cell lines into mature erythrocyte;

passing the differentiated mature erythrocyte through a leukocyte-reducing filter to gather red blood cells; and treating the filtered red blood cell in a low temperature refrigeration, wherein the low temperature is at 0° C. or higher and 15° C. or lower.

20. The method for producing red blood cell of claim **19**, wherein the filtered red blood cell includes nucleated cells; and

wherein the treating in a low temperature refrigeration induces death of the nucleated cells in the filtered red blood cell.

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